

# The Emerging Wonderland Of 'LIVING' COMPUTER SYSTEMS

What if the future of computing is not silicon but living brain cells? Inside laboratories today, scientists are building machines that think, grow, and learn like life itself.



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**L**iving computers, also known as biocomputers, are not yet a part of everyday technology. Like quantum computers, they are still mostly experimental, existing in laboratories and research models. But the idea behind them is powerful: instead of using transistors to process information, biocomputers use living neurons.

Throughout history, humans have found many ways to process information, leading to the dominance of digital computing today. But every time a new method of computing emerges, it brings the potential to change what we can do. Biocomputing is one such unconventional path, offering possibilities that digital systems may never reach. In that way, it is similar to quantum computing, which promises to solve particular problems that digital computers cannot handle.

Biocomputers use much less energy. Neurons are a million times more energy-efficient than silicon for the same computation. Neurons do not grow on a chip—they are grown beforehand in the laboratory. Scaling neurons is easier than silicon because we can multiply them in the laboratory and then

place them on hardware to build a biocomputer. And unlike electronic components, neuron-based systems can be composted at end-of-life, reducing pollution.

There is also a deeper link to artificial intelligence. Most AI today works by simulating neurons, which are called artificial neurons. Systems like ChatGPT and DALL·E run on these digital approximations. But real neurons are far more complex. If we could build AI using real neurons, we might unlock even greater capabilities, while also using far less energy. That is a rare engineering scenario: achieving better performance while simultaneously reducing power consumption.

Of course, biocomputing comes with its own challenges. But the potential gains—in both capability and sustainability—are strong reasons why researchers are still exploring this unusual but promising direction.

## Building the infrastructure for biological computing

Neurons are placed on electrodes to allow two-way communication with electronic hardware. Most of the time, we send electrical signals to the neurons. Chemical signals are the next step, and sometimes we combine electrical and chemical signals to achieve this. One essential component is the use of molecular cages—structures that contain neurotransmitters such as glutamate, serotonin, or dopamine. These cages are placed in the medium surrounding neurons and can be opened with specific wavelengths of light, such as ultraviolet or blue. This enables precise control over chemical release, allowing researchers to mimic brain signalling with high timing accuracy.

A continuous microfluidic system is also required to keep the neurons alive by constantly refreshing the cell culture medium.

